UCB-LSVI: Algorithm, Regret Bound, and Proof

Prepared for class lecture
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Episodic finite-horizon MDP

- Episodes $k = 1, \dots, K$, horizon H. Total time steps $T = KH$.
- State space $\mathcal{S}$, action space $\mathcal{A}$. Rewards $r_h : \mathcal{S} \times \mathcal{A} \to [0, 1]$.
- Transition kernels $P_h(\cdot \mid s, a)$ (possibly time-inhomogeneous).
- Optimal value functions: for $h = 1, \dots, H$,

$$V_h^\star(s) = \max_{a \in \mathcal{A}} Q_h^\star(s, a), \qquad Q_h^\star(s, a) = r_h(s, a) + \mathbb{E}_{s' \sim P_h(\cdot \mid s, a)} \left[V_{h+1}^\star(s') \right],$$

with $V_{H+1}^\star \equiv 0$.

Linear MDP assumption (a.k.a. linear function approximation)

Assumption 1 (Linear MDP; [Jin et al.(2020)Jin, Yang, Wang, and Jordan])

There exists a known feature map $\phi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}^d$ *and unknown vectors* $\{\theta_h\}_{h=1}^H \subset \mathbb{R}^d$ *and measures* $\{\mu_h\}_{h=1}^H$ *over* $\mathcal{S}$ *such that for all* (s, a)

$$r_h(s, a) = \langle \phi(s, a),\, \theta_h \rangle\,,$$
$$P_h(\cdot \mid s, a) = \langle \phi(s, a),\, \mu_h(\cdot) \rangle\,,$$

and $\|\phi(s, a)\|_2 \le 1$, $\|\theta_h\|_2 \le \sqrt{d}$, $\|\mu_h(\mathcal{S})\|_{\mathrm{TV}} \le \sqrt{d}$.

- Implies $Q_h^\star(s, a) = \langle \phi(s, a),\, w_h^\star \rangle$ is linear in ϕ for some $w_h^\star \in \mathbb{R}^d$.
- Dimension d is the *feature dimension* (independent of $|\mathcal{S}|, |\mathcal{A}|$).

Goal: cumulative regret

Definition 2 (Regret)

Let π_k be the policy executed in episode k and $V_1^{\pi_k}(s_1^k)$ its value from the start state s_1^k. The regret after K episodes is

$$\mathrm{Regret}(K) = \sum_{k=1}^{K} \Big(V_1^{\star}(s_1^k) - V_1^{\pi_k}(s_1^k) \Big).$$

When do we fit the value functions?

- The algorithm runs for K episodes, each of horizon H.
- During episode $k-1$, the agent executes policy π_{k-1} and collects data

$$(s_h^i, a_h^i, r_h^i, s_{h+1}^i), \quad h = 1, \ldots, H, \ i = 1, \ldots, k-1.$$

- At the **end of episode** $k-1$, we now have data from the first $k-1$ episodes.
- Before starting episode k, we use *all previously collected data* to **fit new estimates** of the Q-functions and value functions.
- These fitted functions define the new (optimistic) policy π_k that will be used in episode k.

Ridge regression problem in LSVI

Before episode k, we have data from episodes $1, \ldots, k-1$:

$$\mathcal{D}_{h,k-1} = \{(s_h^i, a_h^i, r_h^i, s_{h+1}^i)\}_{i=1}^{k-1} \quad \text{for each stage } h.$$

Ridge regression objective (stage h)

We estimate parameters $w_{h,k}$ by solving

$$w_{h,k} = \arg\min_{w \in \mathbb{R}^d} \sum_{i=1}^{k-1} \Big(r_h(s_h^i, a_h^i) + V_{h+1,k}(s_{h+1}^i) - \langle \phi(s_h^i, a_h^i), w \rangle \Big)^2 + \lambda \|w\|_2^2.$$

- The first term fits observed data and estimated future values.
- The regularization term $\lambda\|w\|^2$ prevents overfitting and ensures invertibility of the design matrix.
- This is a standard **ridge regression** (or L2-regularized least squares) problem.

Ridge regression update equations (before episode k)

For each stage $h = H, H-1, \ldots, 1$,

$$\Lambda_{h,k} = \lambda I + \sum_{i=1}^{k-1} \phi(s_h^i, a_h^i)\phi(s_h^i, a_h^i)^\top,$$

$$w_{h,k} = \Lambda_{h,k}^{-1} \sum_{i=1}^{k-1} \phi(s_h^i, a_h^i)\Big[r_h(s_h^i, a_h^i) + V_{h+1,k}(s_{h+1}^i)\Big].$$

- $\Lambda_{h,k}$ — regularized covariance ("design") matrix.
- $w_{h,k}$ — ridge-regression estimate of the Q-function parameters.
- $Q_{h,k}(s,a) = \langle \phi(s,a), w_{h,k} \rangle, \quad V_{h,k}(s) = \max_a Q_{h,k}(s,a)$.

Optimism via exploration bonuses

- Define the *standard error* at (s, a): $\sigma_{h,k}(s,a) = \sqrt{\phi(s,a)^\top \Lambda_{h,k}^{-1} \phi(s,a)}$.
- UCB-Q estimate:

$$\widehat{Q}_{h,k}(s,a) = \min\Big\{ \langle \phi(s,a),\, w_{h,k} \rangle + \beta_{h,k}\, \sigma_{h,k}(s,a),\; H{-}h{+}1 \Big\}.$$

- Optimistic value: $V_{h,k}(s) = \max_{a \in \mathcal{A}} \widehat{Q}_{h,k}(s,a)$.
- Execute $\pi_k(s) = \arg\max_a \widehat{Q}_{1,k}(s,a)$ in episode k (greedy w.r.t. optimistic values).

UCB-LSVI pseudocode

Input: Horizon H, regularizer $\lambda > 0$, confidence parameter δ, feature map ϕ
Initialize $V_{H+1,1} \equiv 0$ and $\Lambda_{h,1} = \lambda I$ for all h
for $k = 1, 2, \ldots, K$ *episodes* **do**
  for $h = H, H-1, \ldots, 1$ **do**
    Fit $w_{h,k} = \Lambda_{h,k}^{-1} \sum_{i<k} \phi(s_h^i, a_h^i) \left[r_h^i + V_{h+1,k}(s_{h+1}^i) \right]$
    $\sigma_{h,k}(s,a) \leftarrow \sqrt{\phi(s,a)^\top \Lambda_{h,k}^{-1} \phi(s,a)}$
    $\beta_{h,k} \leftarrow$ radius from confidence bound (next slides)
    $\widehat{Q}_{h,k}(s,a) \leftarrow \min\{\langle \phi(s,a),\, w_{h,k} \rangle + \beta_{h,k} \sigma_{h,k}(s,a),\, H-h+1\}$
    $V_{h,k}(s) \leftarrow \max_a \widehat{Q}_{h,k}(s,a)$
  end
  Roll out episode k greedily w.r.t. $\{\widehat{Q}_{h,k}\}_{h=1}^H$
  Update $\{\Lambda_{h,k+1}\}$ with data from episode k
end

Algorithm 1: UCB-LSVI (a.k.a. LSVI-UCB)

Theorem (Regret bound for UCB-LSVI)

Theorem 3 ([Jin et al.(2020)Jin, Yang, Wang, and Jordan])

Fix $\delta \in (0,1)$. Under the linear MDP assumption and with a suitable choice of $\beta_{h,k}$, with probability at least $1-\delta$,

$$\mathrm{Regret}(K) \;\le\; \widetilde{\mathcal{O}}\Big(\sqrt{d^3H^3T}\Big), \qquad T = KH,$$

where $\widetilde{\mathcal{O}}(\cdot)$ hides logarithmic factors in $d, H, T, 1/\delta$.

Proof roadmap (high level)

1. **Self-normalized confidence sets:** For each h, derive a high-probability bound

$$|\langle \phi(s,a),\, w_{h,k} - w_h^\star \rangle| \;\le\; \beta_{h,k}\, \sigma_{h,k}(s,a).$$

2. **Optimism:** Show $\widehat{Q}_{h,k} \ge Q_h^\star$ and hence $V_{h,k} \ge V_h^\star$.
3. **Regret decomposition:** Per-episode regret $\le$ sum of bonuses along visited trajectory.
4. **Bounding cumulative bonuses:** Sum of $\sigma_{h,k}$ terms is controlled by the *elliptical potential lemma*.

Self-normalized bound for linear regression with adaptive data

Let $\{x_t, \eta_t\}$ be a sequence with $\|x_t\| \le 1$, η_t conditionally R-sub-Gaussian. For

$$\Lambda_t = \lambda I + \sum_{i=1}^{t} x_i x_i^\top, \qquad \widehat{\theta}_t = \Lambda_t^{-1} \sum_{i=1}^{t} x_i y_i, \quad y_i = \langle x_i, \theta^\star \rangle + \eta_i,$$

we have w.p. $\ge 1 - \delta$ (Abbasi-Yadkori et al. 2011):

$$\forall x: \ \left| \left\langle x, \widehat{\theta}_t - \theta^\star \right\rangle \right| \le \underbrace{\left(R\sqrt{2 \log \frac{\det(\Lambda_t)^{1/2} \det(\lambda I)^{-1/2}}{\delta}} + \sqrt{\lambda}\, \|\theta^\star\|_2 \right)}_{\beta_t} \sqrt{x^\top \Lambda_t^{-1} x}.$$

Applying to LSVI targets

- For each h, regress targets $y_{h,i} = r_h(s_h^i, a_h^i) + V_{h+1,k}(s_{h+1}^i)$ on $x_i = \phi(s_h^i, a_h^i)$.
- Noise $\eta_{h,i}$ absorbs Bellman residual and next-stage approximation; bounded via induction and clipping to $[0, H-h+1]$.
- Obtain per-layer radii $\beta_{h,k}$ of the form

$$\beta_{h,k} = \mathcal{O}\Big(H\sqrt{\log\frac{\det(\Lambda_{h,k})}{\det(\lambda I)\delta}} + \sqrt{\lambda d}\Big).$$

Optimism via confidence intervals

Lemma 4 (Layer-wise optimism)

Fix an episode k and assume the high-probability confidence event holds: for every h and (s, a),

$$\left|\langle \phi(s,a),\, w_{h,k} - w_h^\star \rangle\right| \;\le\; \beta_{h,k}\sqrt{\phi(s,a)^\top \Lambda_{h,k}^{-1}\phi(s,a)} \;=:\; \beta_{h,k}\,\sigma_{h,k}(s,a).$$

Define the optimistic estimates

$$\widehat{Q}_{h,k}(s,a) \;=\; \min\big\{\langle \phi(s,a), w_{h,k}\rangle + \beta_{h,k}\sigma_{h,k}(s,a),\; H-h+1\big\}, \qquad V_{h,k}(s) = \max_a \widehat{Q}_{h,k}(s,a).$$

Then for all h and s, a,

$$\widehat{Q}_{h,k}(s,a) \;\ge\; Q_h^\star(s,a) \quad \textit{and} \quad V_{h,k}(s) \;\ge\; V_h^\star(s).$$

Proof: Optimism via confidence intervals

Proof.

We prove $\widehat{Q}_{h,k} \geq Q_h^\star$ and hence $V_{h,k} \geq V_h^\star$ by backward induction on h.

Base ($h = H$). Since $V_{H+1}^\star \equiv 0$ and $V_{H+1,k} \equiv 0$, the regression target is $y_{H,i} = r_H(s_H^i, a_H^i)$. On the confidence event,

$$\langle \phi, w_{H,k} \rangle \;\geq\; \langle \phi, w_H^\star \rangle - \beta_{H,k}\sigma_{H,k} \quad \Rightarrow \quad \langle \phi, w_{H,k} \rangle + \beta_{H,k}\sigma_{H,k} \;\geq\; \langle \phi, w_H^\star \rangle \;=\; Q_H^\star.$$

With clipping to $[0, 1] = [0, H - H + 1]$, $\widehat{Q}_{H,k} = \min\{\langle \phi, w_{H,k} \rangle + \beta_{H,k}\sigma_{H,k}, 1\} \geq Q_H^\star$. Thus $V_{H,k}(s) = \max_a \widehat{Q}_{H,k}(s, a) \geq \max_a Q_H^\star(s, a) = V_H^\star(s)$.

Inductive step. Assume for some $h+1 \leq H$ that $V_{h+1,k} \geq V_{h+1}^\star$ pointwise. For any (s, a),

$$\begin{aligned} Q_h^\star(s, a) &= r_h(s, a) + \mathbb{E}_{s' \sim P_h(\cdot|s,a)}\big[V_{h+1}^\star(s')\big] \\ &\leq r_h(s, a) + \mathbb{E}\big[V_{h+1,k}(s')\big] \qquad \text{(by induction hypothesis)} \\ &= \langle \phi(s, a),\, w_h^\star \rangle \qquad \text{(linear MDP: } Q_h^\star(s, a) = \langle \phi, w_h^\star \rangle\text{)} \\ &\leq \langle \phi(s, a),\, w_{h,k} \rangle + \beta_{h,k}\sigma_{h,k}(s, a) \qquad \text{(confidence event).} \end{aligned}$$

Therefore

Per-episode regret is sum of bonuses

Let $\{(s_h^k, a_h^k)\}_{h=1}^H$ be the trajectory under π_k. Optimism gives

$$V_1^\star(s_1^k) - V_1^{\pi_k}(s_1^k) \le V_1^k(s_1^k) - V_1^{\pi_k}(s_1^k) = \sum_{h=1}^H \left(\widehat{Q}_{h,k}(s_h^k, a_h^k) - \left\langle \phi(s_h^k, a_h^k),\, w_{h,k} \right\rangle \right)$$

$$\le \sum_{h=1}^H \beta_{h,k}\, \sigma_{h,k}(s_h^k, a_h^k).$$

Elliptical potential lemma

Lemma 5 (Elliptical potential; e.g., [Abbasi-Yadkori et al.(2011)Abbasi-Yadkori, Pál, and Szepesvári])

Let $\Lambda_0 = \lambda I$ *and* $\Lambda_t = \Lambda_{t-1} + x_t x_t^\top$ *with* $\|x_t\| \leq 1$. *Then*

$$\sum_{t=1}^{T} \min\{1, \sqrt{x_t^\top \Lambda_{t-1}^{-1} x_t}\} \leq \widetilde{\mathcal{O}}\left(\sqrt{d \log \frac{\det(\Lambda_T)}{\det(\Lambda_0)}}\right) \leq \widetilde{\mathcal{O}}\left(\sqrt{d \log \frac{(\lambda + T)^d}{\lambda^d}}\right).$$

Remark 1

In our setting there is one design matrix per layer h, yielding an extra factor $\sqrt{H}$ in the final bound when summing over layers.

Proof of the regret bound

Sketch with scalings.

1. Self-normalized bounds $\Rightarrow$ valid bonuses $\beta_{h,k} = \widetilde{\mathcal{O}}(H\sqrt{d\log(\cdot)})$.
2. Per-episode regret $\leq \sum_{h=1}^{H} \beta_{h,k}\, \sigma_{h,k}(s_h^k, a_h^k)$.
3. Sum over $k = 1..K$ and $h = 1..H$; control $\sum \sigma_{h,k}$ by the elliptical potential lemma for each h.
4. Accounting for H layers and magnitudes of $\beta_{h,k}$ gives

$$\mathrm{Regret}(K) \leq \widetilde{\mathcal{O}}\Big(\sqrt{d^3H^3T}\Big).$$

□

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Confidence radius: a typical form

Under $\|\phi(s,a)\| \le 1$ and rewards in $[0,1]$, a convenient (loose but standard) choice is

$$\beta_{h,k} = c\Big(H\sqrt{d\log\Big(\frac{1+(kH)/\lambda}{\delta}\Big)} + \sqrt{\lambda d}\Big),$$

for a universal constant $c > 0$. Plugging this into the analysis yields the stated $\widetilde{\mathcal{O}}(\sqrt{d^3H^3T})$ regret.

Bounding $\det(\Lambda_T)$

If $\Lambda_T = \lambda I + \sum_{t=1}^{T} x_t x_t^\top$ with $\|x_t\| \le 1$,

$$\log \frac{\det(\Lambda_T)}{\det(\lambda I)} = \sum_{i=1}^{d} \log(1 + \sigma_i/\lambda) \;\le\; d \log\Big(1 + \frac{T}{\lambda d}\Big) \;\le\; \log\Big(\frac{(\lambda + T)^d}{\lambda^d}\Big).$$

Useful inequalities

- $\log(1+x) \le x$; $\sum_{t=1}^{T} \frac{1}{t} \le 1 + \log T$.
- Cauchy–Schwarz; Jensen; Azuma–Hoeffding (for bounded martingale diffs).
- Matrix determinant lemma: $\det(A + uv^\top) = \det(A)(1 + v^\top A^{-1} u)$.

Linear MDP assumption

For each stage h, there exist

$$\theta_h \in \mathbb{R}^d, \qquad \mu_h = (\mu_{h,1}, \ldots, \mu_{h,d})$$

such that for all (s, a):

$$r_h(s, a) = \langle \phi(s, a), \theta_h \rangle, \qquad P_h(\cdot \mid s, a) = \sum_{j=1}^{d} \phi_j(s, a)\, \mu_{h,j}(\cdot).$$

- $\phi(s, a) \in \mathbb{R}^d$ is the feature vector.
- Rewards and transitions are **linear in** ϕ.
- This is the key structural assumption used in LSVI.

Why $Q_h^\star$ is linear in ϕ

The Bellman optimality equation gives

$$Q_h^\star(s,a) = r_h(s,a) + \mathbb{E}_{s' \sim P_h(\cdot|s,a)}[V_{h+1}^\star(s')].$$

Using the linear MDP assumption:

$$r_h(s,a) = \langle \phi(s,a), \theta_h \rangle,$$

$$\mathbb{E}[V_{h+1}^\star(s')] = \sum_{j=1}^{d} \phi_j(s,a) \int V_{h+1}^\star(s')\, \mu_{h,j}(ds').$$

Collecting coefficients:

$$Q_h^\star(s,a) = \Big\langle \phi(s,a),\ \theta_h + \Big(\int V_{h+1}^\star \, d\mu_h \Big) \Big\rangle.$$

Thus:

$$Q_h^\star(s,a) = \langle \phi(s,a),\ w_h^\star \rangle \quad \text{for some } w_h^\star \in \mathbb{R}^d.$$

Structure of the optimal V-function

Given

$$Q_h^\star(s,a) = \langle \phi(s,a), w_h^\star \rangle,$$

the optimal value is

$$V_h^\star(s) = \max_{a \in \mathcal{A}} \langle \phi(s,a), w_h^\star \rangle.$$

- For each fixed a, $Q_h^\star(\cdot, a)$ is **linear in** ϕ.
- $V_h^\star(s)$ is the **maximum of linear functions**.
- Hence $V_h^\star$ is generally **piecewise linear** in feature space.
- Only $Q_h^\star$ needs to lie in the linear span of ϕ for the LSVI analysis.